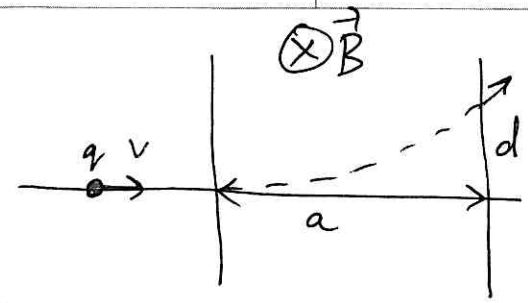
P5.1



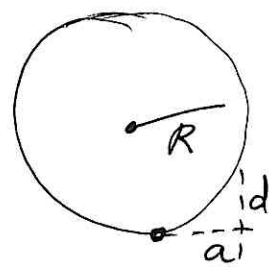
$$\vec{F} = q\,\vec{v}\times\vec{B}$$

$\Rightarrow q>0$ for this geometry.

Magnetic field does no work $\Rightarrow E = \frac{1}{2}mv^2 = \text{const}$

$\Rightarrow |\vec{p}| = m|\vec{v}| = \text{const}.$

Trajectory is segment of a circle



$$qvB = m\frac{v^2}{R}$$

$$R = \frac{mv}{qB} = \frac{p}{qB}$$

$$a^2+d^2 = R^2 \Rightarrow p = qBR = qB\sqrt{a^2+d^2}.$$

P5.2 $y(0) = z(0) = 0$

$$\begin{cases} y(t) = C_1\cos\omega t + C_2\sin\omega t + \frac{E}{B}t + C_3 \\ z(t) = C_2\cos\omega t - C_1\sin\omega t + C_4 \end{cases} ;\ \omega = \frac{qB}{m}$$

$$\left.\begin{aligned} y(0) &= C_1 + C_3 = 0 \\ z(0) &= C_2 + C_4 = 0 \end{aligned}\right\} \Rightarrow \begin{cases} y(t) = C_1(\cos\omega t - 1) + C_2\sin\omega t + \frac{Et}{B} \\ z(t) = C_2(\cos\omega t - 1) - C_1\sin\omega t \end{cases}$$

(a) $\vec{v}(0) = \frac{E}{B}\hat{y} \Rightarrow \dot{y}(0) = \frac{E}{B}$, $\dot{z}(0) = 0$

$$\begin{cases} \dot{y}(t) = -\omega C_1\sin\omega t + \omega C_2\cos\omega t + \frac{E}{B} \\ \dot{z}(t) = -\omega C_2\sin\omega t - C_1\omega\cos\omega t \end{cases}$$

$$\begin{cases} \dot{y}(0) = \omega C_2 + \frac{E}{B} = \frac{E}{B} \Rightarrow C_2 = 0 \\ \dot{z}(0) = -C_1\omega = 0 \Rightarrow C_1 = 0 \end{cases} \Rightarrow C_1 = C_2 = 0$$

$$\begin{cases} z(t) = 0 \\ y(t) = \frac{Et}{B} \end{cases}$$

$\Rightarrow$ this is motion with constant velocity $\vec{v} = \frac{E}{B}\hat{y}$.

(b) $\vec{v}(0) = \frac{E}{2B}\hat{y} \Rightarrow \dot{y}(0) = \frac{E}{2B}, \dot{z}(0) = 0.$

$$\Rightarrow \begin{cases} \omega C_2 + \frac{E}{B} = \frac{E}{2B} \Rightarrow C_2 = -\frac{E}{2B\omega} \\ \omega C_1 = 0 \Rightarrow C_1 = 0 \end{cases}$$

$$\begin{cases} y(t) = -\frac{E}{2B\omega}\sin\omega t + \frac{Et}{B} \\ z(t) = \frac{E}{2B\omega}(1-\cos\omega t) \end{cases}$$

(c) $\vec{v} = \frac{E}{B}(\hat{y}+\hat{z}) \Rightarrow \dot{y}(0) = \frac{E}{B}, \dot{z}(0) = \frac{E}{B}$

$$\omega C_2 + \frac{E}{B} = \frac{E}{B} \Rightarrow C_2 = 0$$

$$-\omega C_1 = \frac{E}{B} \Rightarrow C_1 = -\frac{E}{B\omega}$$

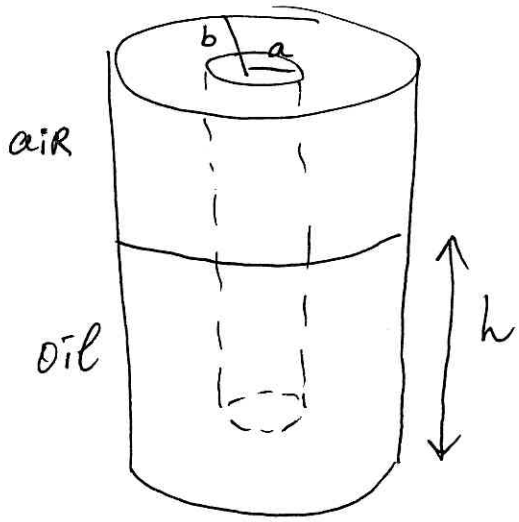
$$\begin{cases} y(t) = +\frac{E}{\omega B}(1-\cos\omega t) + \frac{Et}{B} \\ z(t) = \frac{E}{B\omega}\sin\omega t \end{cases}$$

P5.3 (a) zero deflection requires $\vec{v} = \frac{E}{B}\hat{y}$ as we saw in solving P5.2(a)

$$\Rightarrow v = \frac{E}{B}$$

(b) $qvB = m\frac{v^2}{R} \Rightarrow R = \frac{mv}{qB} \Rightarrow \frac{q}{m} = \frac{v}{BR} = \frac{E}{R}$.

P4.28

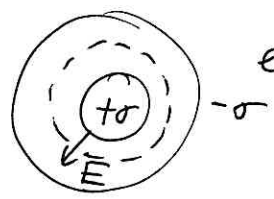


Electrostatic force on dielectric liquid is $F = \frac{1}{2}V^2\frac{dC}{dh}$.

It is equilibrated by gravitational $F_g = mg$

where $m = \rho\,\pi(b^2 - a^2)\,h$ is the mass of dielectric liquid.

$\Rightarrow$ find $C(h)$:



empty capacitor first:

$$E \cdot 2\pi r l = \frac{2\pi a l \sigma}{\varepsilon_0}$$

$$E = \frac{\sigma a}{\varepsilon_0 r}$$

$$V = -\int_a^b E\,dr = -\frac{\sigma a}{\varepsilon_0}\int_a^b \frac{dr}{r} = \frac{\sigma a}{\varepsilon_0}\ln\frac{b}{a}$$

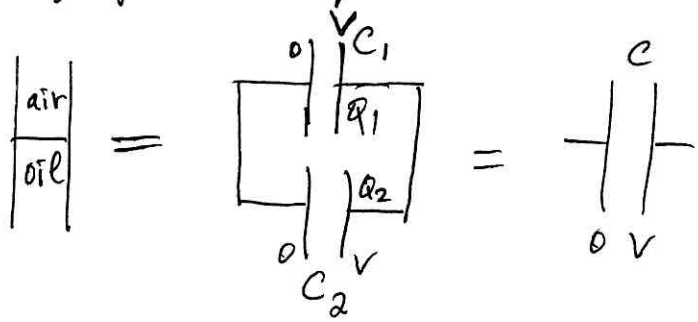
$$\sigma = \frac{Q}{\text{Area}} = \frac{Q}{2\pi a l}$$ where l = length of the cylinders.

$$\Rightarrow C_{\text{empty}} = \frac{Q}{V} = \frac{2\pi\varepsilon_0 l}{\ln(b/a)}$$ for empty capacitor.

In the fully filled capacitor $E = \frac{D}{\varepsilon} = \frac{\sigma a}{\varepsilon r} \Rightarrow C_{\text{filled}} = \frac{2\pi\varepsilon_0(1+\chi_e)l}{\ln(b/a)}$

$\varepsilon = \varepsilon_0\varepsilon_r = \varepsilon_0(1+\chi_e)$

Partially filled capacitor = two capacitors in parallel



$\Rightarrow C = C_1 + C_2$

$$Q = VC = Q_1 + Q_2 = VC_1 + VC_2$$

$$\Rightarrow C = C_1 + C_2$$

$$C_1 = \frac{2\pi\varepsilon_0 (\ell - h)}{\ln(b/a)} \; ; \; C_2 = \frac{2\pi\varepsilon_0 (1+\chi_e) h}{\ln(b/a)}$$

$$C = \frac{2\pi\varepsilon_0}{\ln(b/a)} \cdot \left[\ell - h + (1+\chi_e) h\right]$$

$$\Rightarrow \frac{dC}{dh} = \frac{2\pi\varepsilon_0 \chi_e}{\ln(b/a)}$$

$$\Rightarrow \frac{1}{2} V^2 \frac{2\pi\varepsilon_0 \chi_e}{\ln(b/a)} = \pi(b^2 - a^2)\rho g h$$

$$h = \frac{\varepsilon_0 \chi_e V^2}{(b^2 - a^2)\ln\left(\frac{b}{a}\right)\rho g}$$

P4.40 $U = -\vec{p}\cdot\vec{E} \Rightarrow \langle U\rangle = -\langle\vec{p}\rangle\cdot\vec{E}$ where $\langle\vec{p}\rangle$ is the average dipole moment.

let $\vec{E} = E\hat{x}$

In terms of $\vec{p}\cdot\vec{E} = pE\cos\theta$, $\langle\vec{p}\rangle = \langle\cos\theta\rangle\hat{x}$.

Find average orientation via

$$\langle\cos\theta\rangle = \frac{\int \cos\theta\, e^{-U/k_BT}\, d\varphi \sin\theta d\theta}{\int e^{-U/k_BT}\, d\varphi \sin\theta d\theta}.$$

Nothing depends on $\varphi \Rightarrow \int d\varphi = 2\pi$ and cancels out.

$$\langle\cos\theta\rangle = \frac{\int_0^\pi \cos\theta\, e^{\frac{pE\cos\theta}{k_BT}} \sin\theta\, d\theta}{\int_0^\pi e^{\frac{pE}{k_BT}\cos\theta} \sin\theta d\theta} = (x = \cos\theta) =$$

$$= \frac{\int_{-1}^{1} e^{\frac{pE}{k_BT}x}\, x\, dx}{\int_{-1}^{1} e^{\frac{pEx}{k_BT}} dx}$$

let $\frac{pE}{k_BT} = s \Rightarrow$ we need $I_1 = \int_{-1}^{1} e^{sx} dx = \frac{1}{s}\int_{-s}^{s} e^{y} dy =$

$$= \frac{e^{s} - e^{-s}}{s} = 2\frac{sh(s)}{s}.$$

Then $I_2 = \int_{-1}^{1} e^{sx}\, x dx = \frac{d}{ds}\int_{-1}^{1} e^{sx} dx = \frac{d}{ds}\left(\frac{2\,sh(s)}{s}\right) =$

$$= 2\left(\frac{ch(s)}{s} - \frac{sh(s)}{s^2}\right). = \frac{2}{s}\, sh(s)\left(\frac{ch(s)}{sh(s)} - \frac{1}{s}\right).$$

Thus $\langle \cos\theta \rangle = \frac{I_2}{I_1} = \left(\coth(s) - \frac{1}{s} \right)$, where $s = \frac{pE}{k_B T}$.

$$\Rightarrow \langle p \rangle = p \langle \cos\theta \rangle$$

Polarization per unit volume $P = N \langle p \rangle$

$$\Rightarrow P = Np \left(\coth\left(\frac{pE}{k_B T}\right) - \frac{k_B T}{pE} \right).$$

(b) For $s = \frac{pE}{k_B T} \ll 1$ we can expand the result

Need $\coth(s) = \frac{\cosh(s)}{\sinh(s)} = \frac{1 + \frac{1}{2}s^2}{s + \frac{s^3}{6}} = \frac{1}{s}\left(1 + \frac{s^2}{2}\right)\left(1 - \frac{s^2}{6}\right) =$

$$= \frac{1}{s}\left(1 + \left(\frac{1}{2} - \frac{1}{6}\right)s^2 + \dots\right) = \frac{1}{s} + \frac{s}{3} + \dots$$

$$\Rightarrow P = Np \left(\frac{k_B T}{pE} + \frac{1}{3}\frac{pE}{k_B T} - \frac{k_B T}{pE} \right) = \frac{Np^2 E}{3 k_B T}$$

$P = \alpha E \Rightarrow \alpha = \frac{Np^2}{3 k_B T}$ at high temperature.